

Classification- Accuracy

		Predicted condition	
Total population = P + N		Positive (PP)	Negative (PN)
Actual condition	Positive (P)	True positive (TP), hit	False negative (FN), type II error, miss, underestimation
	Negative (N)	False positive (FP), type I error, false alarm, overestimation	True negative (TN), correct rejection

$$\text{Accuracy} = (\text{TP} + \text{TN}) / (\text{TP} + \text{FN} + \text{FP} + \text{TN})$$

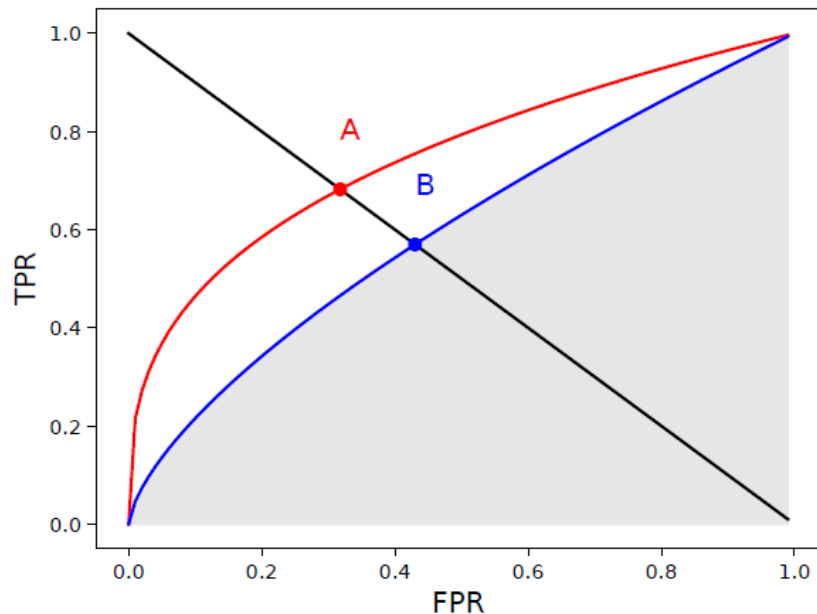
Recall / True Positive Rate (TPR) = predict how many detected correctly in cohort

$$\text{TPR} = \text{TP} / (\text{TP} + \text{FN})$$

False Positive Rate (FPR) = predict how many detected wrongly in cohort

$$\text{FPR} = \text{FP} / (\text{FP} + \text{TN})$$

Classification- ROC curve



EER: $FPR = FNR = 1 - TPR$

Adapted from <https://probml.github.io/pml-book/teaching1.html> by Prof. Saw Shier Nee

Area under ROC curve is computed.

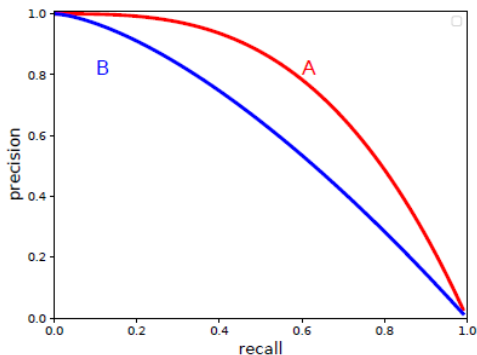
The higher the AUROC, the better the classifier.

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$TPR = TP / (TP + FN) \rightarrow$ predict how many disease correctly in disease cohort

$FPR = FP / (FP + TN) \rightarrow$ predict how many disease in healthy cohort

Classification- Precision-Recall Curve

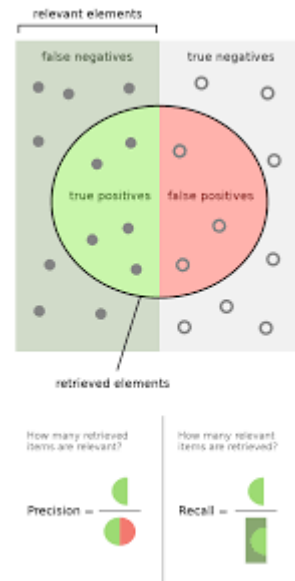


Recall = TPR = $TP / (TP + FN)$ → how many predicted items are correct among all relevant items
Precision = Positive Predictive Value (PPV) = $TP / (TP + FP)$ → how many predicted items are correct among all retrieved items.

Area under PR curve is computed.

The higher the AUC of PR curve, the better the classifier.

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F1 score: Harmonic mean of Precision and Recall (encourages similar values of precision and recall)

Cumulative Match Characteristic

A Cumulative Match Characteristic (CMC) curve plots the probability of identification against the returned 1:N candidate list size.

It shows the probability that a given user appears in different sized candidate lists. The faster the CMC curve approaches 1, indicating that the user always appears in the candidate list of specified size, the better the matching algorithm.